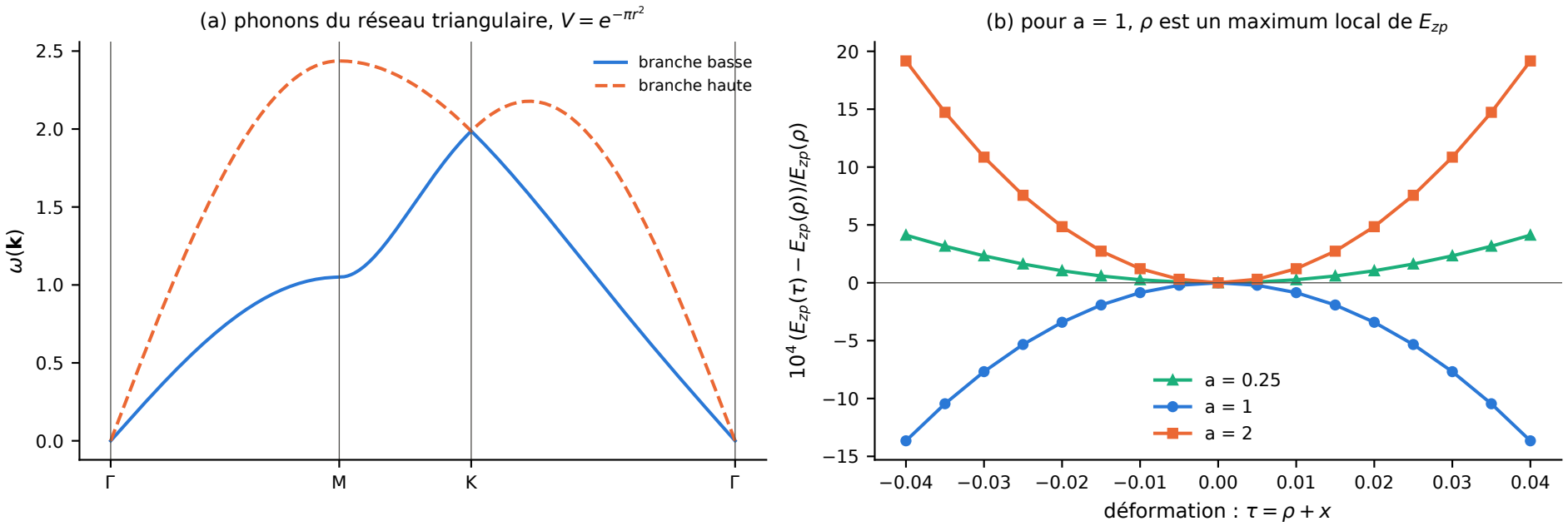
(a) phonons du réseau triangulaire, $V = e^{-\pi r^2}$
branche basse
branche haute
$\omega(\mathbf{k})$
2.5
2.0
1.5
1.0
0.5
0.0
Γ
M
K
Γ
(b) pour a = 1, ρ est un maximum local de E_{zp}
$10^4\,(E_{zp}(\tau) - E_{zp}(\rho))/E_{zp}(\rho)$
20
15
10
5
0
−5
−10
−15
a = 0.25
a = 1
a = 2
−0.04
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
déformation : $\tau = \rho + x$